

MODULE: AI FOR MEDICAL DEVICE

Project Report:

Machine Learning-Based Classification of Glioma Using TCGA Dataset

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Abstract

One of the broad categories of brain tumours that are primary in nature are the gliomas, which can be further divided into low-grade glioma (LGG) and glioblastoma multiforme (GBM). Proper classification can be crucial in the prognosis and choice of treatment strategies. This paper examines the use of machine learning method to classify glioma with the use of The Cancer Genome Atlas (TCGA) database.

A machine learning pipeline was created and had been organized in such a way that data preprocessing and feature engineering took place, model training was done and an evaluation was performed. Several classification models such as Logistic Regression, Support Vector Machines (SVM), Random Forest, Gradient Boosting and AdaBoost were compared on the basis of stratified cross-validation. L1-regularisation and ensemble learning employing soft voting classifier were used as features selection to improve model performance. A voting ensemble with a high discriminating power is a soft voting ensemble whose performance is highest with the accuracy of 0.876 and AUC is 0.927. Sensitivity/specificity trade-off was also observed in the threshold tuning, vital in clinical decision making.

Evaluation of the importance of features indicated that the IDH1 mutation status and age at diagnosis is one of the significant predictors. These findings suggest the potential of using machine learning techniques to help clinicians in their diagnosis and decision-making in the classification of glioma.

1. Introduction

Gliomas are the most common and complex types of primary brain tumours. Their origin is in glial cells and they are highly heterogeneous in behaviour, progression and patient outcomes. Glioma is divided into low-grade glioma (LGG) and high-grade (i.e., glioblastoma multiforme (GBM)). GBM is extremely aggressive and with low survival rates and LGG is slow progressing with high prognosis.

Grading of glioma properly is quite essential in planning of treatment that involves surgery, radiotherapy and chemotherapy. Traditional methods of diagnosis that might be time-consuming and responsive to inter-observer variation have been histopathological analysis of tumour tissue (Naser and Deen, 2020). This has drawn more focus to the use of computational methods to assist and enhance the quality of diagnosis. With the availability of large genomic data sets, such as The Cancer Genome Atlas (TCGA), machine learning has become a powerful tool to analyse the complex biological data. Using machine learning models, it is possible to extract high-dimensional data trends to predict and classify them to a greater degree. However, such pitfalls as loss of data, redundant features, and overfitting are to be avoided to obtain credible performance. The research will assist in developing a powerful machine learning pipeline to classify glioma using TCGA data. The objectives are to assess the performance of different classification models, apply feature selection and ensemble learning algorithms and to provide a clinical meaning to the results.

2. Methods

2.1 Dataset

The variable of interest, Grade is tumour typology according to the low-grade glioma (LGG) and glioblastoma multiforme (GBM). After preprocessing the dataset, 861 samples and 23 predictive features were used. Data employed in this research was accessed in the cancer Genomic Atlas (TCGA) that gives detailed genomic and clinical information on cancer studies. The data includes details of patients with glioma, including demographic factors (gender, race, age at the time of diagnosis) and binary factors (e.g., IDH1, TP53, ATRX, EGFR) that reflect whether a patient has a gene mutation or not (Liñares-Blanco, Pazos and Fernandez-Lozano, 2021). Grade is a target variable that implies the classification of tumours as low-grade glioma

(LGG) or glioblastoma multiforme (GBM). After preprocessing, the data was 861 samples and 23 predictive variables.

2.2 Data Preprocessing

Preprocessing of the data was undertaken to make the data consistent, eliminate noises, and underscore the data to be used in machine learning modelling. The first indication of non-standard missing values (Not Reported, etc.) has been replaced with NaN values. This standardisation allowed the handling of missing data in the next processing phases accordingly. The Age at diagnosis feature was initially a text field (e.g. 51 years 108 days). This variable was coded into a continuous numerical variable reflecting the age in years to enable it to be utilized by machine learning algorithms. Removed irrelevant and non-informative predictors, such as CaseID, Project, etc., as they are identifiers but not predictors (Tasci et al., 2022). Further, the optional feature PrimaryDiagnosis was not used to avoid leakage of data because it is an immediate relationship with the target variable as it can artificially inflate the model performance should it be kept.

2.3 Feature Engineering, and Transformation.

The variables included in the data were categorical and numerical. It employed proper preprocessing methods in order to be compatible with machine learning models. The median value was used to impute the numerical features so that the missing values could be handled and then standardised with StandardScaler. Standardisation is used to give all the numeric features similar scales, this is of great importance to algorithms that are sensitive to the size of features, e.g. Support Vector Machines. Most frequent category was used to impute categorical features, and one-hot encoded (Yuan et al., 2024). This change puts the categorical variables in form of binary that can be used in machine learning models. As a precautionary method to reduce vulnerability upon cross-validation, the encoder was set to handle unknown= ignore, such that the model can easily tolerate the unknown categories in the validation data without any error.

2.4 Model Development

Supervised machine learning models were tested based on a variety of algorithms to compare their performance based on classification. The models included:

- Linear Model (probabilistic output, logistic regression)

- Probability estimation Support Vector Machine (SVM).
- Random Forest (ensemble trees model)
- Gradient Boosting (sequential ensemble model)
- AdaBoost (adaptive boosting model)

Such models were chosen to represent two approaches to the classification, namely linear and non-linear classification, which offer an overall assessment of the performance.

2.5 Cross-Validation

Stratified 5-fold cross-validation was used to measure model performance. This method will be used to split the dataset into five folds and at the same time maintain the balance of LGG and GBM cases within a fold. Stratified cross-validation helps to minimize sampling bias and assure that the measures of performance represent the entire dataset (Lu et al., 2018). This is of particular importance in medical datasets where the imbalance of classes can be a problem.

2.6 Feature Selection

L1-regularised Logistic Regression (Lasso) was used in the process of feature selection. This approach involves a penalty that causes less significant feature coefficients to diminish to zero and eliminates redundant or confounding features. This method enhances the interpretability of the model, minimises overfitting, and maximises the performance of generalisation by narrowing in on the most significant predictors.

2.7 Ensemble Learning

A soft voting ensemble classifier was introduced to further enhance the predictive performance. This model takes predictions of various base classifiers (Logistic Regression, SVM and Random Forest) and predicts using an average of the predicted probabilities. Ensemble learning reduces model variance and improves the robustness by exploiting the performance of a large number of algorithms. The soft voting was chosen over hard voting since it does not just use the labels of the classes, but also takes into account the confidence in prediction.

2.8 Evaluation Metrics

- Various evaluation measures were used to evaluate model performance and give a analyzing picture:
- Accuracy: Percentage of correct forecasts in general.
- Precision: Percentage of positive cases that are predicted right.
- Recall (Sensitivity): The capability to find GBM cases correctly.
- F1-score: Precision and recall ratioed to each other case wise.
- Specificity: Capacity to appropriately note LGG cases.
- AUC (Area Under the ROC Curve): that of the total classification performance over thresholds.

Recall is particularly beneficial in clinical practice since it decreases the false negative rate and ensures that problematic GBM cases are not missed. Specificity is another critical aspect that will help to avoid treating LGG patients inappropriately.

2.9 Threshold Tuning

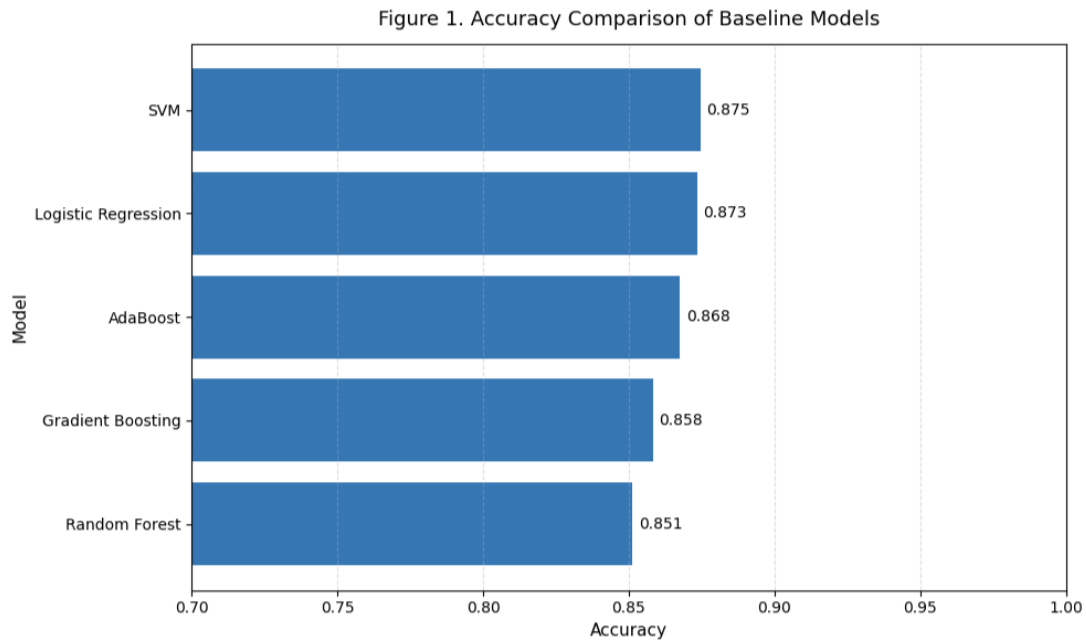
Also threshold tuning was included as well as the standard classification evaluation so that the impact of decision thresholds can be studied on the model performance. The default threshold was (0.5) with a range of 0.1 to 0.85. The thresholds at the lower end are more sensitive (recall) and result in more GBM cases, but higher thresholds are more specific (lower false positives). This discussion will provide a clue on how the model can be customized to different clinical priorities.

3. Results

3.1 Baseline Model Performance

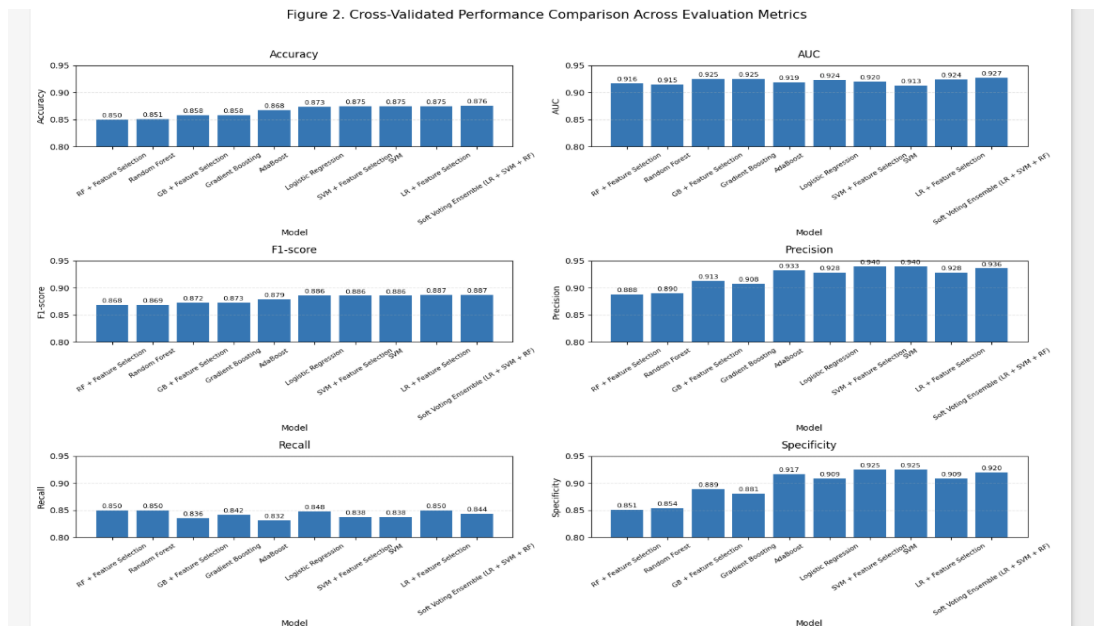
The results of the baseline machine learning models are summarised in Figure 1. It was discovered that Support Vector machine (SVM) had the highest accuracy of approximately 0.875 after which the Logistic regression followed with an accuracy of 0.873. Random Forest, Gradient Boosting and AdaBoost also did not fail miserably with an accurateness score of above 0.84. The implications of these findings are that the data can be linearly separated to

some extent with linear models of Logistic Regression and SVM especially performing well. However, non-linear ensemble models also yielded good results, implying the existence of complex feature interaction.



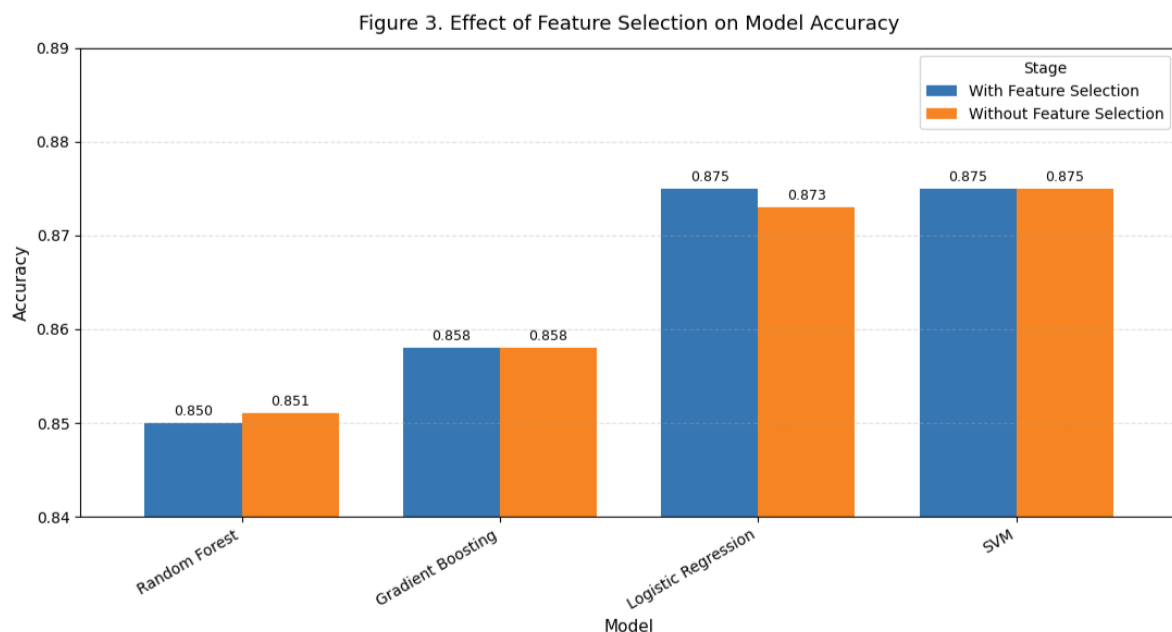
3.2 Feature Selection Results

The feature selection reduced the predictive performance, without any impact on the model complexity. L1-regularised Logistic Regression achieved a comparable accuracy of about 0.875 that is slightly lower than the baseline model. This means that a considerable portion of original characteristics was unnecessary or vaguely informative (Yuan et al., 2024). This ability to maintain performance using minimum number of features suggests that the classification task is driven by the lower number of highly predictiveness features. Not only does it enhance the interpretability of the model it also helps to minimise the chances of overfitting, increasing the likelihood of generalisability to unknown data.



3.3 Ensemble Model Performance

The best overall performance was found in the soft voting ensemble, with an accuracy of 0.876 and an AUC of 0.927, and a high precision (0.936) and an evenly distributed recall (0.844). The overall excellent performance in all analysis measures reveals that the ensemble is effective in capturing complementary individual model observed patterns. This enhancement can be explained by decreased model variance and increased generalisation, which prove that using multiple classifiers results in stronger predictions and dependability in comparison to using a single model.



3.4 Threshold Tuning

The trade-off between specificity, overall classification balance, and recall of varying the decision threshold were analysed. Figure 4 shows the result

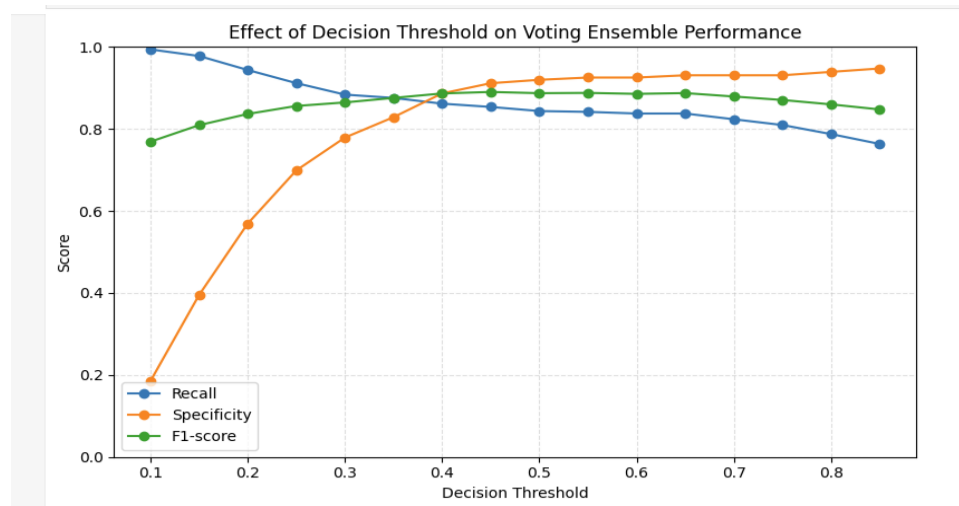


Figure 4. Influence of decision threshold on voting ensemble decisionmaking. The plot shows the variation of recall, specificity and F1-score with the threshold, showing how the two conflict in terms of sensitivity and precision.

As demonstrated in Figure 4, when the decision threshold becomes smaller, the amount of recall increases significantly, the closer it gets to 1.0. This means a high sensitivity with almost all the GBM cases being correctly identified. This is, however, at the expense of lowering specificity, which increases the amount of false positives. On the other hand, specificity increases with threshold better reaching a high of 0.94 with a higher threshold, but the recall decreases. This looks like a decrease in false positives and a bigger chance of under reporting actual GBM cases. The F1-score is similar at the middle ranges (around 0.35-0.65) indicating that the range offers an optimal trade-off between accuracy and recall. This is especially in the clinical applications where false negative and false positive are both life-threatening. On the whole, these findings prove that the performance of the models can be aligned with the priorities in the clinical setting. A smaller threshold can be desirable in high-risk diagnostic situations to maximise sensitivity, but a more critical threshold can be desirable in situations where a reduction of false positives is essential (Liñares-Blanco, Pazos and Fernandez-Lozano, 2021).

3.5 ROC Curve Analysis

A Receiver Operating Characteristic (ROC) curve was also used to further evaluate the discriminative performance of the final model, as depicted in Figure 5.

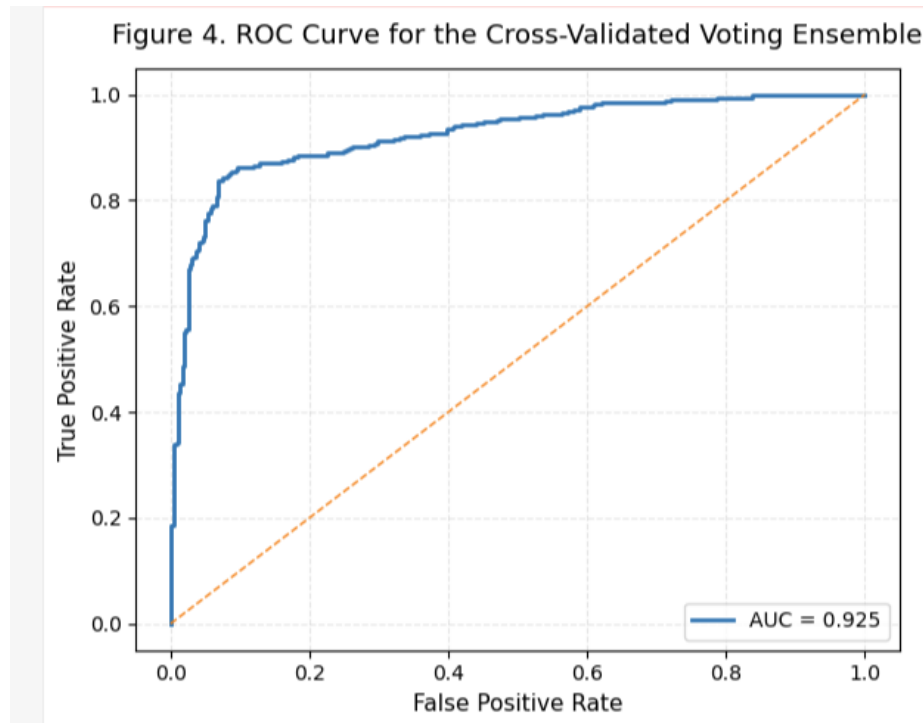


Figure 5. The cross-validated soft voting ensemble model ROC curve.

The AUC of the model is about 0.925, which means that the model is very efficient in distinguishing between LGG and GBM cases. As Figure 5 demonstrates, the ROC curve is very far over the reference line of the diagonal, which means that the model is much better than random classification. The extreme slope of the curve at low false positive rates shows that the model is very sensitive as well as the false positive rates are relatively low. This is indicated by the high value of AUC ($=0.925$), which reveals the high-level discriminative ability of the model meaning that at a variety of decision levels, the model works very well in the classification of low-grade and high-grade glioma cases.

3.6 Feature Importance

Analysis of the power of features analysed the coefficients of final model with embedded features selection. Figure 6 shows the best predictors

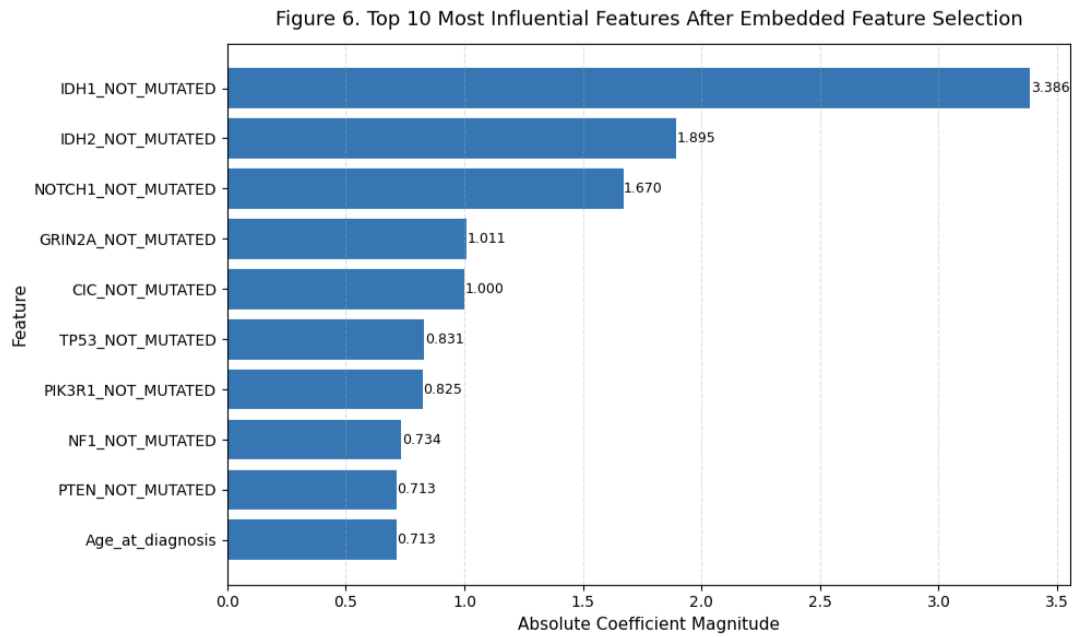


Figure 6. Top 10 influential features following embedded feature selection.

The feature importance is calculated on the basis of the absolute of the feature in the model coefficients, which gives the contribution of any feature to the classification task as compared to the other feature. IDH1 mutation status turns out as the most significant predictor stemming as the largest coefficient value than all the other features, as illustrated in Figure 6. This underscores its predominance in separating LGG vs. GBM cases. The IDH2 mutation and NOTCH1, further boosted with other genomic markers including GRIN2A, CIC and TP53. The age of diagnosis is one of the highest features but with a relatively low contribution, meaning that, although it is a clinically relevant factor, genomic markers are still a stronger classification factor (Tasci et al., 2022). The general trend indicates that key genetic changes are the main stimuli of this model and not demographic factors. Such results are in line with known clinical experience, in which IDH mutations have been identified as important biomarkers to classify and prognose glioma. Biological consistency (agreement with known biology) and alignment of model outputs grounds validity and interpretability of the model.

4. Discussion

The findings allow concluding that with the help of genomic and clinical features, machine learning models can be effectively used to classify the types of glioma. The good results of SVM and Logistic Regression indicate that the dataset is properly organized and can be used

during the classification. The best performance was obtained with the soft voting ensemble which means that the optimization of multiple models predictability is better. This holds with other literature which has shown that ensemble techniques are in most cases superior to single models (Yuan et al., 2024). The feature importance analysis showed that the most important predictor is the IDH1 mutation. It agrees with clinical literature that a significant biomarker of glioma classification and prognosis is IDH1. Its clinical relevance was also in terms of age at diagnosis. Clinically, high recall is crucial to reduce false negatives and prevent GBM cases without detection. Nevertheless, there is also the need to have high specificity in order to prevent unnecessary treatment to LGG patients. This trade-off is shown by the threshold tuning results and shows the ability to vary model performance according to clinical demands. In spite of good performance, there are a number of constraints. The amount of data is quite average, and it was not externally checked (Tasci et al., 2022). One-hot encoding can also induce sparsity, which can affect the performance of the model. Future research might consider larger datasets, deep learning representations, and clinical validation.

5. Conclusion

This piece of work had developed a more comprehensive machine learning pipeline that classifies glioma using TCGA data. The results indicate that feature selection and ensemble learning would be able to improve the model performance, yet, feature selection and ensemble learning maintain interpretability. The clinically relevant factors, such as IDH1 mutation status and age are identified in support of the biological validity of the model. Though the outcomes require further validation, they suggest that machine learning strategies can be used as a tool to support clinical decision-making in healthcare.

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